

A novel der(12)t(7;12)(p15;q24.3) in a patient with childhood T-cell acute lymphoblastic leukemia.

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Approximately 35% of T-cell acute lymphoblastic leukemia (T-ALL) cases have chromosomal translocations as evaluated by conventional cytogenetic methods (G-banding). Some chromosomal translocations are associated with morphologically and immunophenotypically distinct leukemia subtypes and define patients with different clinical outcomes. Chromosomal translocations may deregulate gene expression, thus contributing to the development of neoplasia, either by placing a putative oncogene under the control of strong regulatory elements or by generating chimeric genes and oncogenic fusion proteins. We report here a novel der(12)t(7;12)(p15;q24.3) in a child with T-ALL. Cloning and characterization of the breakpoint region may contribute to the discovery of new genes that are important in T-ALL.

Epigenomic translocation of H3K4me3 broad domains over oncogenes following hijacking of super-enhancers.

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Chromosomal translocations are important drivers of haematological malignancies whereby proto-oncogenes are activated by juxtaposition with enhancers, often called enhancer hijacking. We analyzed the epigenomic consequences of rearrangements between the super-enhancers of the immunoglobulin heavy locus (IGH) and proto-oncogene CCND1 that are common in B cell malignancies. By integrating BLUEPRINT epigenomic data with DNA breakpoint detection, we characterized the normal chromatin landscape of the human IGH locus and its dynamics after pathological genomic rearrangement. We detected an H3K4me3 broad domain (BD) within the IGH locus of healthy B cells that was absent in samples with IGH-CCND1 translocations. The appearance of H3K4me3-BD over CCND1 in the latter was associated with overexpression and extensive chromatin accessibility of its gene body. We observed similar cancer-specific H3K4me3-BDs associated with hijacking of super-enhancers of other common oncogenes in B cell (MAF, MYC, and FGFR3/NSD2) and T cell malignancies (LMO2, TLX3, and TAL1). Our analysis suggests that H3K4me3-BDs can be created by super-enhancers and supports the new concept of epigenomic translocation, in which the relocation of H3K4me3-BDs from cell identity genes to oncogenes accompanies the translocation of super-enhancers.

MYC Drives a Subset of High-Risk Pediatric Neuroblastomas and Is Activated through Mechanisms Including Enhancer Hijacking and Focal Enhancer Amplification.

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The amplified MYCN gene serves as an oncogenic driver in approximately 20% of high-risk pediatric neuroblastomas. Here, we show that the family member MYC is a potent transforming gene in a separate subset of high-risk neuroblastoma cases (~10%), based on (i) its upregulation by focal enhancer amplification or genomic rearrangements leading to enhancer hijacking, and (ii) its ability to transform neuroblastoma precursor cells in a transgenic animal model. The aberrant regulatory elements associated with oncogenic MYC activation include focally amplified distal enhancers and translocation of

highly active enhancers from other genes to within topologically associating domains containing the MYC gene locus. The clinical outcome for patients with high levels of MYC expression is virtually identical to that of patients with amplification of the MYCN gene, a known high-risk feature of this disease. Together, these findings establish MYC as a bona fide oncogene in a clinically significant group of high-risk childhood neuroblastomas. Significance: Amplification of the MYCN oncogene is a recognized hallmark of high-risk pediatric neuroblastoma. Here, we demonstrate that MYC is also activated as a potent oncogene in a distinct subset of neuroblastoma cases through either focal amplification of distal enhancers or enhancer hijacking mediated by chromosomal translocation. *Cancer Discov*; 8(3); 320-35. ©2017 AACR. This article is highlighted in the In This Issue feature, p. 253.

Altered enhancer-promoter interaction leads to MNX1 expression in pediatric acute myeloid leukemia with t(7;12)(q36;p13).

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Acute myeloid leukemia (AML) with the t(7;12)(q36;p13) translocation occurs only in very young children and has a poor clinical outcome. The expected oncofusion between break point partners (motor neuron and pancreas homeobox 1 [MNX1] and ETS variant transcription factor 6 [ETV6]) has only been reported in a subset of cases. However, a universal feature is the strong transcript and protein expression of MNX1, a homeobox transcription factor that is normally not expressed in hematopoietic cells. Here, we map the translocation break points on chromosomes 7 and 12 in affected patients to a region proximal to MNX1 and either introns 1 or 2 of ETV6. The frequency of MNX1 overexpression in pediatric AML is 2.4% and occurs predominantly in t(7;12)(q36;p13) AML. Chromatin interaction assays in a t(7;12)(q36;p13) induced pluripotent stem cell line model unravel an enhancer-hijacking event that explains MNX1 overexpression in hematopoietic cells. Our data suggest that enhancer hijacking may be a more widespread consequence of translocations in which no oncofusion product was identified, including t(1;3) or t(4;12) AML.

Oncogenic Enhancers in Leukemia.

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Although the study of leukemogenesis has traditionally focused on protein-coding genes, the role of enhancer dysregulation is becoming increasingly recognized. The advent of high-throughput sequencing, together with a better understanding of enhancer biology, has revealed how various genetic and epigenetic lesions produce oncogenic enhancers that drive transformation. These aberrations include translocations that lead to enhancer hijacking, point mutations that modulate enhancer activity, and copy number alterations that modify enhancer dosage. In this review, we describe these mechanisms in the context of leukemia and discuss potential therapeutic avenues to target these regulatory elements. Significance: Large-scale sequencing projects have uncovered recurrent gene mutations in leukemia, but the picture remains incomplete: some patients harbor no such aberrations, whereas others carry only a few that are insufficient to bring about transformation on their own. One of the missing pieces is enhancer dysfunction, which only recently has emerged as a critical driver of leukemogenesis. Knowledge of the various mechanisms of enhancer dysregulation is thus key for a complete understanding of leukemia and its causes, as well as the development of targeted therapies in the era of

precision medicine.

HYENA detects oncogenes activated by distal enhancers in cancer.

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Somatic structural variations (SVs) in cancer can shuffle DNA content in the genome, relocate regulatory elements, and alter genome organization. Enhancer hijacking occurs when SVs relocate distal enhancers to activate proto-oncogenes. However, most enhancer hijacking studies have only focused on protein-coding genes. Here, we develop a computational algorithm 'HYENA' to identify candidate oncogenes (both protein-coding and non-coding) activated by enhancer hijacking based on tumor whole-genome and transcriptome sequencing data. HYENA detects genes whose elevated expression is associated with somatic SVs by using a rank-based regression model. We systematically analyze 1146 tumors across 25 types of adult tumors and identify a total of 108 candidate oncogenes including many non-coding genes. A long non-coding RNA TOB1-AS1 is activated by various types of SVs in 10% of pancreatic cancers through altered 3-dimensional genome structure. We find that high expression of TOB1-AS1 can promote cell invasion and metastasis. Our study highlights the contribution of genetic alterations in non-coding regions to tumorigenesis and tumor progression.

Enhancer hijacking activates GFI1 family oncogenes in medulloblastoma.

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Medulloblastoma is a highly malignant paediatric brain tumour currently treated with a combination of surgery, radiation and chemotherapy, posing a considerable burden of toxicity to the developing child. Genomics has illuminated the extensive intertumoral heterogeneity of medulloblastoma, identifying four distinct molecular subgroups. Group 3 and group 4 subgroup medulloblastomas account for most paediatric cases; yet, oncogenic drivers for these subtypes remain largely unidentified. Here we describe a series of prevalent, highly disparate genomic structural variants, restricted to groups 3 and 4, resulting in specific and mutually exclusive activation of the growth factor independent 1 family proto-oncogenes, GFI1 and GFI1B. Somatic structural variants juxtapose GFI1 or GFI1B coding sequences proximal to active enhancer elements, including super-enhancers, instigating oncogenic activity. Our results, supported by evidence from mouse models, identify GFI1 and GFI1B as prominent medulloblastoma oncogenes and implicate 'enhancer hijacking' as an efficient mechanism driving oncogene activation in a childhood cancer.

Histone acetylation dynamics modulates chromatin conformation and allele-specific interactions at oncogenic loci.

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In cancer cells, enhancer hijacking mediated by chromosomal alterations and/or increased deposition of acetylated histone H3 lysine 27 (H3K27ac) can support oncogene expression. However, how the chromatin conformation of enhancer-promoter interactions is affected by these events is unclear. In the

present study, by comparing chromatin structure and H3K27ac levels in normal and lymphoma B cells, we show that enhancer-promoter-interacting regions assume different conformations according to the local abundance of H3K27ac. Genetic or pharmacological depletion of H3K27ac decreases the frequency and the spreading of these interactions, altering oncogene expression. Moreover, enhancer hijacking mediated by chromosomal translocations influences the epigenetic status of the regions flanking the breakpoint, prompting the formation of distinct intrachromosomal interactions in the two homologous chromosomes. These interactions are accompanied by allele-specific gene expression changes. Overall, our work indicates that H3K27ac dynamics modulates interaction frequency between regulatory regions and can lead to allele-specific chromatin configurations to sustain oncogene expression.

Enhancer hijacking activates oncogenic transcription factor NR4A3 in acinic cell carcinomas of the salivary glands.

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The molecular pathogenesis of salivary gland acinic cell carcinoma (AciCC) is poorly understood. The secretory Ca-binding phosphoprotein (SCPP) gene cluster at 4q13 encodes structurally related phosphoproteins of which some are specifically expressed at high levels in the salivary glands and constitute major components of saliva. Here we report on recurrent rearrangements [t(4;9)(q13;q31)] in AciCC that translocate active enhancer regions from the SCPP gene cluster to the region upstream of Nuclear Receptor Subfamily 4 Group A Member 3 (NR4A3) at 9q31. We show that NR4A3 is specifically upregulated in AciCCs, and that active chromatin regions and gene expression signatures in AciCCs are highly correlated with the NR4A3 transcription factor binding motif. Overexpression of NR4A3 in mouse salivary gland cells increases expression of known NR4A3 target genes and has a stimulatory functional effect on cell proliferation. We conclude that NR4A3 is upregulated through enhancer hijacking and has important oncogenic functions in AciCC.

Could the HALP score serve as a biomarker of bronchiectasis exacerbation?

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Bronchiectasis is a chronic inflammatory lung disease and patients may occasionally experience acute exacerbations. Our study aims to determine the relationship between exacerbation periods and HALP (hemoglobin, albumin, lymphocyte, platelet) scores in patients with bronchiectasis. Adult patients diagnosed with bronchiectasis and followed up in our clinic between 02.2020-12.2022 were retrospectively evaluated. After the examinations, the effect of bronchiectasis exacerbation on the HALP score was investigated. A total of 84 patients diagnosed with non-cystic fibrosis bronchiectasis were included in our study. 42 of the patients were male (50%), and 42 were female. The average age of all patients was 52.37 ± 16.2 . 35 patients (41.7%) were in the exacerbation period, and 49 patients (58.3%) were in the stable period. The median values of leukocytes, neutrophils, and C-reactive protein (CRP) were significantly higher in patients during the exacerbation period compared to the stable period (respectively $p = 0.00$, $p = 0.00$, $p = 0.00$). The average values of FEV1% and FVC% in patients during the exacerbation period were significantly lower compared to the stable period ($p = 0.03$, $p = 0.00$, respectively). The HALP score was significantly lower in patients during the exacerbation period compared to the stable period ($p = 0.00$). A significant negative correlation was found between the HALP

score and leukocytes, neutrophils, and CRP ($p = 0.00$, $p = 0.00$, $p = 0.00$, respectively). Also, a significant positive correlation was found between the HALP score and FEV1% and FVC% ($p = 0.00$, $p = 0.00$, respectively). Our study revealed that the HALP score is associated with infectious and pulmonary functional parameters in bronchiectasis patients in the exacerbation period. We propose that the HALP score could serve as a valuable biomarker during exacerbations.
